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EXAMPLE

$X \sim \Gamma(\alpha, \lambda)$, $Y \sim \Gamma(\beta, \lambda)$, where $\alpha, \beta, \lambda > 0$ and X, Y indep.
What is the dist of $X+Y$?

Solution

We know $f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,dx$

$$(z>0) = \int_0^z \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)} \cdot \frac{\lambda^\beta (z-x)^{\beta-1} e^{-\lambda(z-x)}}{\Gamma(\beta)}\,dx$$

$$= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda z} \int_0^z x^{\alpha-1}(z-x)^{\beta-1}\,dx$$

Let $x = zu$,

$$= \frac{\lambda^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda z} \int_0^1 (zu)^{\alpha-1}(z-zu)^{\beta-1} z\,du$$

$$= \frac{\lambda^{\alpha+\beta} z^{\alpha+\beta-1}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\lambda z} \int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,du$$

$$= \frac{\lambda^{\alpha+\beta} z^{\alpha+\beta-1} e^{-\lambda z}}{\Gamma(\alpha)\Gamma(\beta)} \cdot C_{\alpha,\beta}$$ for some const $C_{\alpha,\beta}$ $\Rightarrow$ proportional to $f_Z(z)$ for $Z \sim \Gamma(\alpha+\beta, \lambda)$

However, the integral over $\mathbb{R}$ has to equal 1!
$\therefore$ $X+Y \sim \Gamma(\alpha+\beta, \lambda)$

SUMMARY

If $X \sim \Gamma(\alpha, \lambda)$ and $Y \sim \Gamma(\beta, \lambda)$ are indep, then $X+Y \sim \Gamma(\alpha+\beta, \lambda)$

COROLLARY

If $X_i \sim \underbrace{\text{Exp}(\lambda)}_{\sim \Gamma(1,\lambda)}$, $1 \le i \le n$, are indep, then $\sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$

EXAMPLE

Let $Z_i \sim N(0,1)$, $1 \le i \le n$, be indep. What is the distribution of $\sum_{i=1}^n Z_i^2$?

Solution

What is the distribution of Z_i^2?

CDF: $F_{Z_i^2}(a) = \mathbb{P}(Z_i^2 \le a) = \mathbb{P}(-\sqrt{a} \le Z_i \le \sqrt{a}) = \int_{-\sqrt{a}}^{\sqrt{a}} f_{Z_i}(z)\,dz = \int_{-\sqrt{a}}^{\sqrt{a}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,dz = 2\int_0^{\sqrt{a}} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}\,dz$

$\therefore$ $f_{Z_i^2}(a) = \frac{d}{da} F_{Z_i^2}(a) = \frac{1}{\sqrt{2\pi}} e^{-\frac{a}{2}} a^{\frac{1}{2}-1}$ (FTC)

Notice, $Y \sim \Gamma(\alpha, \lambda) \Rightarrow f_Y(a) = \frac{\lambda^\alpha a^{\alpha-1} e^{-\lambda a}}{\Gamma(\alpha)}$

$\therefore$ $Z_i^2 \sim \Gamma(\frac{1}{2}, \frac{1}{2})$

$\therefore$ $\sum_{i=1}^n Z_i^2 \sim \Gamma(\frac{n}{2}, \frac{1}{2})$

Rmk: This is the chi-squared dist w/ df=n.

QUESTION

Let $X_i \sim N(\mu_i, \sigma_i^2)$ be indep, $1 \le i \le n$. What is the dist of $\sum_{i=1}^n X_i$?

PROPOSITION

$\sum_{i=1}^n X_i \sim N(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)$

Proof

Plan:

① Show $n=2 \Rightarrow$ general case (ind on n)
② Show that $\mu_1 = \mu_2 = 0$, $\sigma_1 = 1$, $\sigma_2 = \sigma \Rightarrow$ general case
③ Prove $\mu_1 = \mu_2 = 0$, $\sigma_1 = 1$, $\sigma_2 = \sigma$ case

Actual proof:

① Proof by ind on n,
Induction step: $X_1 + \ldots + X_n + X_{n+1} = \underbrace{(X_1 + \ldots + X_n)}_{N(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2)} + \underbrace{X_{n+1}}_{N(\mu_{n+1}, \sigma_{n+1}^2)}$, indep
By $n=2$ case, $X_1 + \ldots + X_{n+1} \sim N(\sum_{i=1}^n \mu_i + \mu_{n+1}, \sum_{i=1}^n \sigma_i^2 + \sigma_{n+1}^2) = N(\sum_{i=1}^{n+1} \mu_i, \sum_{i=1}^{n+1} \sigma_i^2)$ ✓

② Base Case ($n=2$)

$X_1 \sim N(\mu_1, \sigma_1^2)$; $X_2 \sim N(\mu_2, \sigma_2^2)$

Then, $X_1+X_2 = \sigma_1\left(\frac{X_1+X_2}{\sigma_1}\right) = \sigma_1\left(\frac{X_1-\mu_1}{\sigma_1} + \frac{X_2-\mu_2}{\sigma_1} + \frac{\mu_1+\mu_2}{\sigma_1}\right) = \sigma_1\left(\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1}\cdot\frac{X_2-\mu_2}{\sigma_2}\right) + (\mu_1+\mu_2)$

Notice, $\frac{X_1-\mu_1}{\sigma_1} \sim N(0,1)$ and $\frac{\sigma_2}{\sigma_1}\frac{X_2-\mu_2}{\sigma_2} \sim N(0, \frac{\sigma_2^2}{\sigma_1^2})$

If $\mu_1=\mu_2=0$, $\sigma_1=1$, $\sigma_2=\sigma$ case is true, then $\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1}\cdot\frac{X_2-\mu_2}{\sigma_2} \sim N(0, 1+\frac{\sigma_2^2}{\sigma_1^2})$

$\therefore \sigma_1\left(\frac{X_1-\mu_1}{\sigma_1} + \frac{\sigma_2}{\sigma_1}\frac{X_2-\mu_2}{\sigma_2}\right) \sim N(0, \sigma_1^2+\sigma_2^2)$

$\therefore X_1+X_2 \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$ ✓

③ $X_1 \sim N(0,1)$, $X_2 \sim N(0,\sigma^2)$, indep

Let $X = X_1+X_2$

We know $f_{X_1}(t) = \frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}$, $f_{X_2}(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{x^2}{2\sigma^2}}$

$\therefore f_X(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy$

$= \frac{1}{2\pi\sigma}\int_{-\infty}^{\infty} e^{-\frac{(z-y)^2}{2} - \frac{y^2}{2\sigma^2}} dy$ (Exponent $= -\frac{(z^2-2zy+y^2)}{2} - \frac{y^2}{2\sigma^2} = -\frac{z^2}{2} + \frac{2zy\sigma^2 - y^2\sigma^2 - y^2}{2\sigma^2}) = -\frac{z^2}{2} - \frac{\sigma^2+1}{2\sigma^2}\left(y - \frac{2\sigma^2 zy}{\sigma^2+1}\right) = -\frac{z^2}{2} - \frac{\sigma^2+1}{2\sigma^2}\left(y - \frac{\sigma^2}{\sigma^2+1}z\right)^2 + \frac{\sigma^2}{2(\sigma^2+1)}z^2$)

$= \frac{1}{2\pi\sigma} e^{-\frac{z^2}{2} + \frac{\sigma^2}{2(\sigma^2+1)}z^2} \int_{-\infty}^{\infty} e^{-\frac{\sigma^2+1}{2\sigma^2}\left(y - \frac{\sigma^2}{2(\sigma^2+1)}z\right)^2} dy$

Let $y = 1 - \frac{\sigma^2}{\sigma^2+1}z$, $= \frac{1}{2\pi\sigma} e^{-\frac{z^2}{2} + \frac{\sigma^2}{2(\sigma^2+1)}z^2} \underbrace{\int_{-\infty}^{\infty} e^{-\frac{\sigma^2+1}{2\sigma^2}u^2} du}$ ← some constant

$= C' e^{-\frac{(\sigma^2+1)z^2 - z^2\sigma^2}{2(\sigma^2+1)}} = C' e^{-\frac{z^2}{2(\sigma^2+1)}}$

$\therefore$ This is the PDF of $N(0, 1+\sigma^2)$ □

DEFINITION

We say a var X is log-normally distributed if $\log X$ is normal

EXAMPLE

There is a stock whose price after n weeks is S_n.

The ratios $\frac{S_n}{S_{n-1}}$ are indep and log-normal with $\mu = 0.0165$, $\sigma = 0.0730$

① What is the probability that $S_n > S_{n-1}$ for the 52 weeks of a year?

② What is the probability that $S_n > S_{n-1}$ for at least 30 weeks in the year?

③ What is the probability that the stock will have gained value over the course of the year?

Solution

Notice, $\{S_n > S_{n-1}\} = \{\frac{S_n}{S_{n-1}} > 1\} = \{\log\frac{S_n}{S_{n-1}} > 0\}$, where $\log\frac{S_n}{S_{n-1}} \sim N(0.0165, 0.0730^2)$

Let $X \sim N(0.0165, 0.0730^2)$.

Then, $\mathbb{P}(X>0) = \mathbb{P}\left(\frac{X-0.0165}{0.0730} > -\frac{0.0165}{0.0730}\right) = \mathbb{P}\left(Z \leq \frac{0.0165}{0.0730}\right) = 0.5894$

① $\mathbb{P}$(gaining val all 52 weeks)

$= \mathbb{P}(\{S_1 > S_0\} \cap \{S_2 > S_1\} \cap \ldots \cap \{S_{52} > S_{51}\})$

$= \mathbb{P}(\{\log\frac{S_1}{S_0} > 0\} \cap \ldots \cap \{\log\frac{S_{52}}{S_{51}} > 0\})$

$= \mathbb{P}(\log\frac{S_1}{S_0} > 0) \cdots \mathbb{P}(\log\frac{S_{52}}{S_{51}} > 0)$ (independence)

$= 0.5894^{52} \approx 1.15 \times 10^{-12}$

② Let Y = #weeks in the year with $S_n > S_{n-1}$

$\therefore Y \sim Bin(52, 0.5894)$ by indep

Using Normal Approx, $Y \sim N(52 \times 0.5894, 52 \times 0.5894 \times (1-0.5894))$

$\therefore \mathbb{P}(Y \geq 30) = \mathbb{P}(Y \geq 29.5) \simeq \mathbb{P}\left(\frac{Y - 52(0.5894)}{\sqrt{52(0.5894)(1-0.5894)}} \geq \frac{29.5 - 52(0.5894)}{\sqrt{52(0.5894)(1-0.5894)}}\right) \approx 0.627$

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③ $$\mathbb{P}(S_{52} > S_0) = \mathbb{P}\left(\log\left(\frac{S_{52}}{S_0}\right) > 0\right) = \mathbb{P}\Bigg(\underbrace{\sum_{i=1}^{52} \log\left(\frac{S_n}{S_{n-1}}\right)}_{\substack{\sim \\ N(52\times 0.0165,\ 52\times 0.0730^2)}} > 0\Bigg)$$

$\therefore\ \mathbb{P}(S_{52} > S_0) = 0.9484$